

CHARLENE CHAO-RONG CHEN

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EDUCATION

University College London

2023 – 2027

Master of Engineering Robotics and Artificial Intelligence

London, United Kingdom

First Class Honours (4.0 cumulative GPA)

Relevant Coursework: Linear Algebra, Multivariable Calculus, Discrete Maths, Probability and Statistics, Optimisation, Differential Equations, Data Structures and Algorithms, Mechatronics, Control System, Machine Learning, Modelling and Simulation

WORK EXPERIENCE

Machine Learning Engineer Intern

June 2025 – September 2025

Autodesk

London, United Kingdom

- Designed and implemented research optimising data embedding for reasoning capabilities in generative 3D models
- Achieved 3.5x improvement in embedding efficiency while maintaining 80%+ model accuracy using PyTorch and AWS services

AI Engineering Intern

March 2025 – June 2025

Cisco

London, United Kingdom

- Built multi-agent LLM system (LangGraph, Claude, OpenAI) with tool use over IoT printer APIs, converting natural language to physical prototypes in <15 mins
- Deployed containerised agentic workflows in Python/TypeScript, reducing setup time 95% and accelerating prototyping 4x

Software Engineering Research Intern

June 2024 – September 2024

AcoustoFab, UCL Interaction Centre Multisensory-Devices Group

London, United Kingdom

- Implemented acoustic manipulation for high-precision sorting on phased arrays of transducers using C++ and machine vision
- Project featured by UCL Engineering: "Ultrasound technology could transform agricultural seed sorting"

Assistant Data Analysis Engineer

June 2022 – July 2022

Siemens

Taipei, Taiwan

- Analysed 4 months of Siemens' energy and public datasets to identify efficiency drivers and forecast infrastructure usage

LEADERSHIP EXPERIENCE

President

June 2025 – Present

UCL Artificial Intelligence Society

London, United Kingdom

- Lead six society initiatives, including AI Foundry, student-led research lab, weekly machine learning tutorials, company speaker sessions, and social networking events

PROJECTS

Tactile Representation Learning for Embodied AI

February 2026 – Present

UCL CRISP Lab (Cognitive Robotics & Intelligent Systems), supervised by Prof. Lorenzo Jamone

London, United Kingdom

- Empirical research on tactile representation learning for embodied agents, focused on data quality and downstream generalisation
- Built ROS2-based system for synchronised collection of contact-rich sensor data to construct controlled datasets

Curiosity-Driven Self-Touch Learning in Embodied Agents

April 2026

Independent Research Project

London, United Kingdom

- Showed balanced intrinsic/extrinsic rewards yield richer body representations, training curiosity-driven agents with PPO-based Random Network Distillation in BabyBench
- Revealed purely reward-driven policies exploit sensor shortcuts, evaluated via DreamerV3 world model quality

Security Challenges in Decentralized Multi-Agent Reinforcement Learning (MARL)

March 2026

Independent Literature Review

London, United Kingdom

- Analysed attack surfaces across MARL systems (data poisoning, policy-level manipulation), showing how centralized critics amplify local corruption into system-wide policy failures
- Evaluated reward-hacking and adversarial-testing mitigations, identifying open problems in decentralised agent security

SKILLS

Languages: English (Native), Chinese (Native)

Programming Languages: Python, C, Java, MATLAB

Tools/Frameworks: PyTorch, Scikit-Learn, LangGraph, Gymnasium/MuJoCo, ROS2, Ray, AWS, Docker, Linux, Git

Certifications: NVIDIA-Certified Associate for Generative AI LLMs